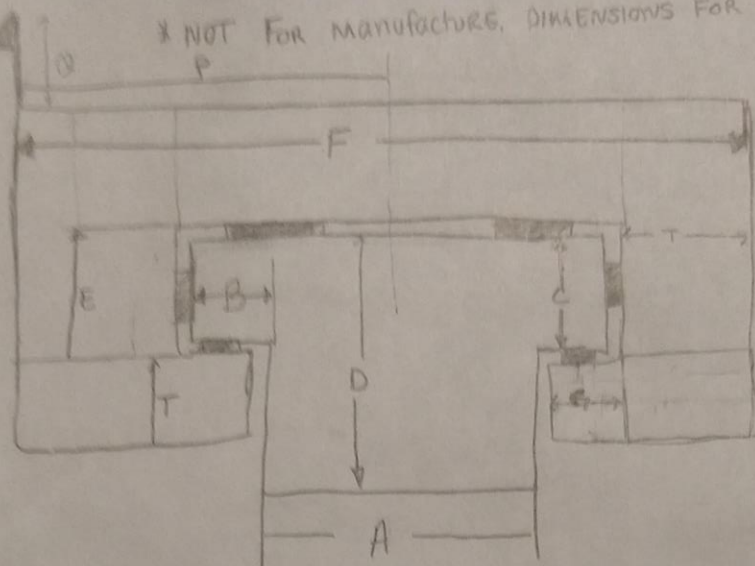
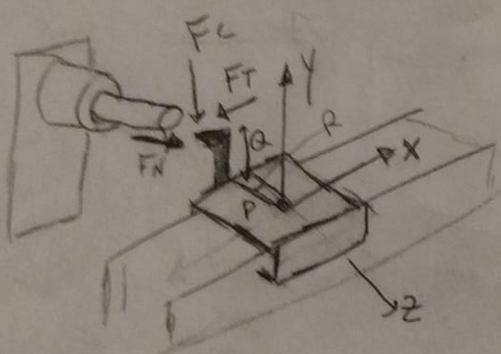
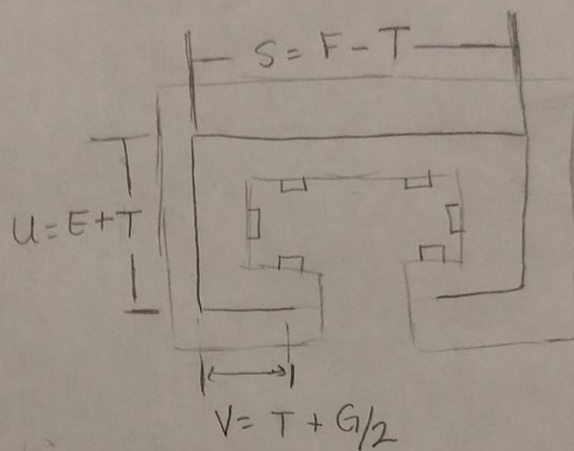
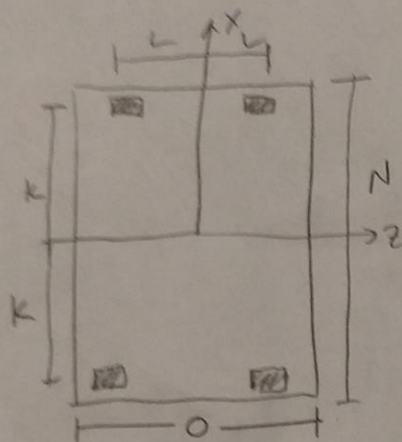


Tool → * NOT FOR MANUFACTURE, DIMENSIONS FOR DESIGN PURPOSES



$$F - 2t > (A + 2B)$$

$$F - 2t - 2G > A$$

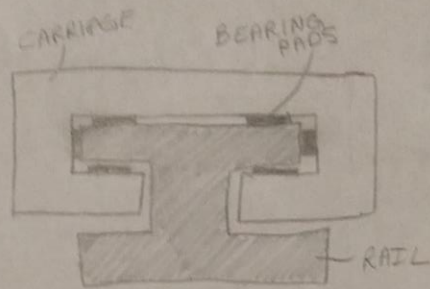
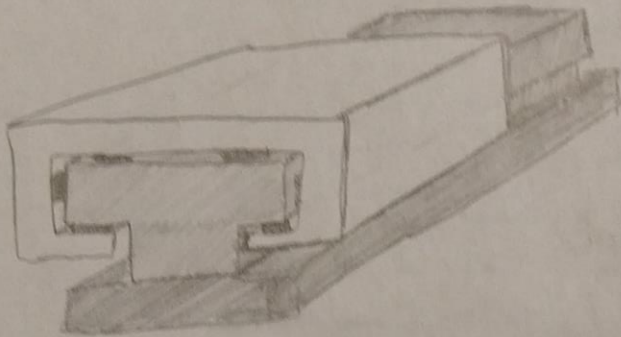


$$\sum M_x = -F_c \cdot P + F_N \cdot Q$$

$$\sum M_y = F_T \cdot P - F_N \cdot R$$

$$\sum M_z = F_T \cdot Q - F_c \cdot R$$

Boxway Error Motion Derivations



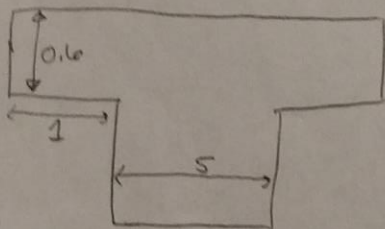
- FIRST ORDER APPROXIMATION OF BEARING PAD STIFFNESS



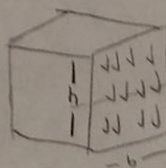
$$K_{pad} = \frac{AE}{t} = \frac{(LW)E}{t}$$

→ can be used for bearing pad and rubber pad (spring preload)

→ Want T-Rail to act in shear AND NOT as a BENDING BEAM.
 ↳ use St. Venants to get a good starting point for proportions



→ now will not bend as a beam (too chunky). it only acts in shear (Typically negligible but you can easily calculate it).



V = shear force

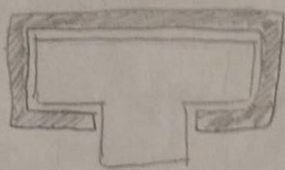
$$\tau = \frac{VQ}{It} = \frac{V(\frac{1}{2}(\frac{h^2}{4} - y^2)b)}{\frac{bh^3}{12} \cdot b}$$

$$= \frac{6V}{bh^3}(\frac{h^2}{4} - y^2) = 1.5 \frac{V}{A} = \tau_n$$

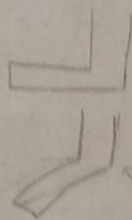
@ neutral ax

$$\tau_{avg} = \frac{V}{A}$$

→ NEED TO ENSURE CARRIAGE DOES Act in Beam bending in the appropriate places

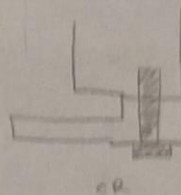


=>

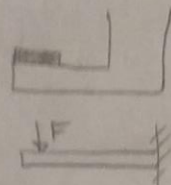


is this how it bends?
all the stiffness is in the corner.

Give it a diving board =>

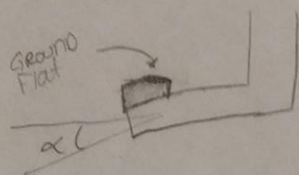


or



now it actually behaves like a cantilevered BEAM

* Be aware of WEAR!



Cutting FORCE (Kalpakjian & Schmid Manufacturing Engineering and Tech.)

Total allowable ERROR

0.1mm seems reasonable. For system

→ allowable load induced error

axis error apportionment

Bearings : 13.27 μm

Structure : 6.63 μm

Total = 19.9 μm
But we have 3 axes
so

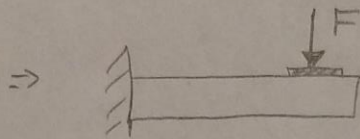
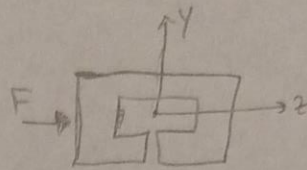
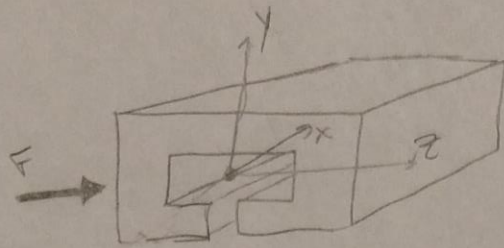
desired deflection for one axis is

$$19.9/3 = 6.63$$

δ_{max}

$$K_{\text{LMS}} = F_{\text{cut}} / \delta_{\text{max}} = 24 \text{ N}/\mu\text{m} = 24,000 \text{ N}/\text{mm}$$

Z-dir stiffness

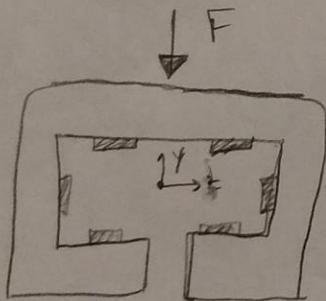


$$K_{\text{structure}} = \frac{F}{\left(\frac{FL^3}{3EI}\right)}$$

$$K_{\text{eq}} = \left(\frac{1}{\frac{1}{K_{\text{structure}}} + \frac{1}{K_{\text{pad}}}} \right)$$

$$\delta_{\text{el total}} = \frac{F}{K_{\text{eq}}}$$

Y-dir stiffness



=>

Not really a cantilever beam in this case.

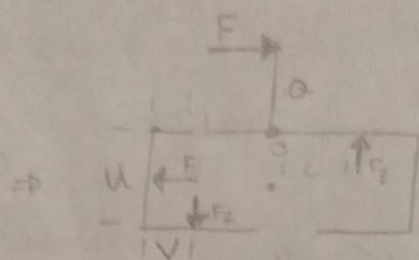
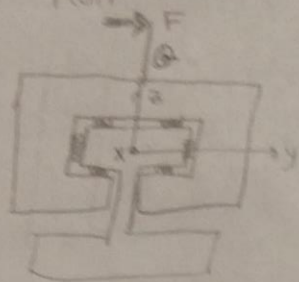
only deflection will come from the compression of the bearing pad and the compression of the carriage and rail

$$K_{\text{eq}} = \left(\frac{1}{\frac{1}{K_{\text{carriage}}} + \frac{1}{K_{\text{bear}}}} \right)$$

$$K_{\text{carriage}} = \frac{E_c A_c}{t_c}$$

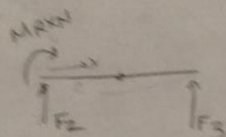
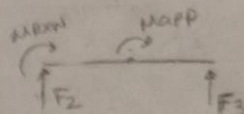
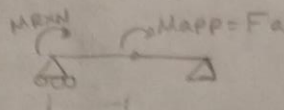
$$K_{\text{pad}} = \frac{E_p A_p}{t_{\text{pad}}}$$

1. Roll Stiffness

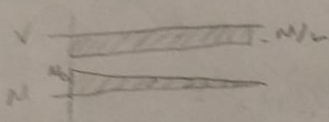


* Rxns act at center of bearing pads

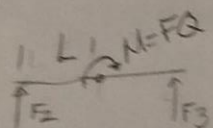
$$M_{RXN} = F_1 \left(\frac{U}{2} \right) + F_2 V$$



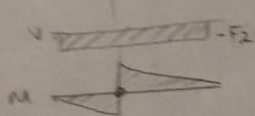
$$\delta = \frac{-M(U)}{6EI} (2U^2 - 3L(U) + L^2)$$



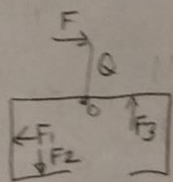
$$\theta = \frac{-M}{6EI} (2U^2 - 6L(U) + 3L^2)$$



$$\theta_2 = \frac{-M}{6EI(2L)} (2U^2 - 6L(U) + 3L^2) + \left(\frac{-M}{2L} \right) \frac{X^2}{2EI} + \frac{M_0}{EI} \langle X-L \rangle'$$



$$\delta = \frac{-M}{6EI(2L)} (2U^2 - 6L(U) + 3L^2) \times \left(\frac{M}{2L} \right) \frac{X^3}{6EI} + \frac{M}{2EI} \langle X-L \rangle^2$$

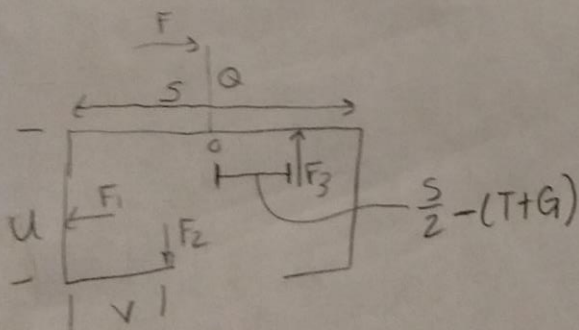


$$\sum F_z = 0 \Rightarrow F = F_1$$

$$\sum F_y = 0 \Rightarrow F_3 = F_2$$

$$\sum M_0 = 0 \Rightarrow FQ - F_3 \left(\frac{S}{2} - (T+G) \right) + F_1 \left(\frac{U}{2} \right) - F_2 \left(\frac{S}{2} - V \right) = 0$$

$$F_3 = \frac{F_1 \left(Q + \frac{U}{2} \right)}{\left(\left(\frac{S}{2} - V \right) + \frac{S}{2} - (T+G) \right)}$$

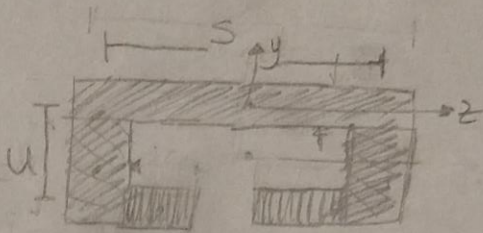
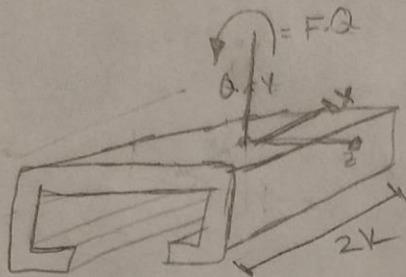


$$M_{RXN} = F_1 \left(\frac{U}{2} \right) + F_2 (V)$$

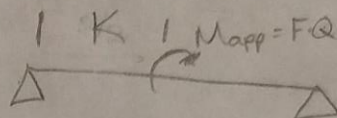
$$\Rightarrow F \left(\frac{U}{2} \right) + \frac{F \left(Q + \frac{U}{2} \right)}{\left(\left(\frac{S}{2} - V \right) + \frac{S}{2} - (T+G) \right)} \cdot V$$

$$K_{roll} = \frac{M_{app}}{(\theta_1 + \theta_2)}$$

Pitch stiffness



→ Find I_{yy}

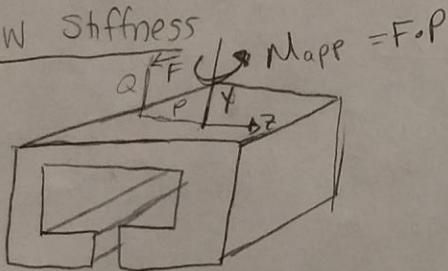


I_{yy} for Ibeam approximation
Quick!

$$K_{pitch} = \frac{M_{app}}{\theta_z} \rightarrow F \cdot Q$$

$$\theta_z = \frac{-M}{6EI} \left(2(2k)^2 - 6(k)(2k) + 3(k^2) \right) + \frac{\left(\frac{-M}{2k} \right) \left(\frac{x^2}{2k} \right)}{2EI} + \frac{M}{EI} \langle x - k \rangle$$

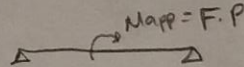
Yaw stiffness



⇒



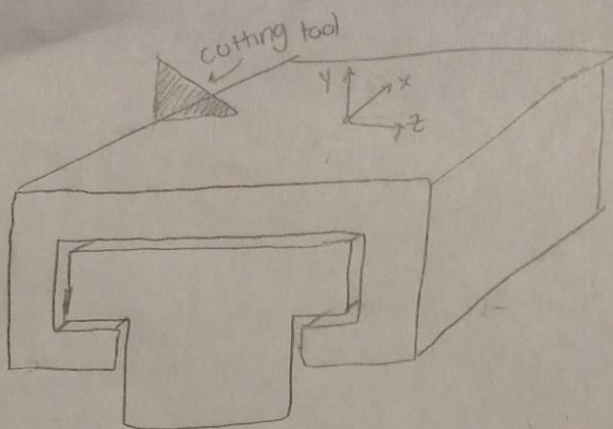
→ Find I_{yy} . ⇒ $I = I_{Ibeam}$ for quick approximation



$$K_{pitch} = \frac{M_{app}}{\theta_z}$$

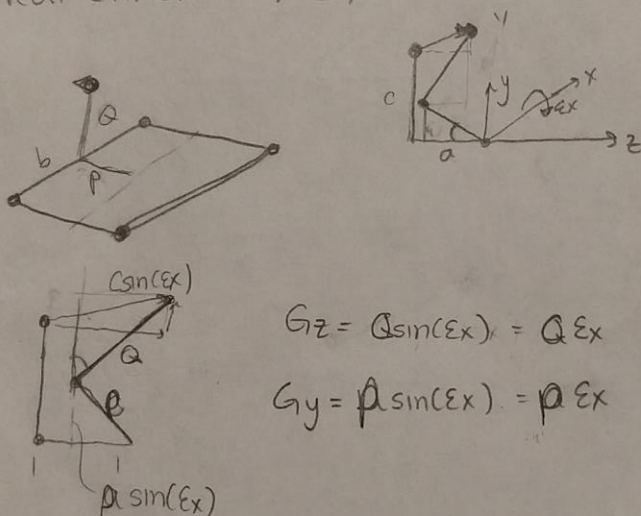
$$\theta_y = \frac{\left(\frac{-M}{6EI} \right) \left(2(2k)^2 - 6k(2k) + 3(k^2) \right) + \frac{\left(\frac{-M}{2k} \right) \left(\frac{x^2}{2k} \right)}{2EI} + \frac{M}{EI} \langle x - k \rangle$$

$$K_{yaw} = \frac{M_{app}}{\theta_y}$$



Calculating Error Gain matrix

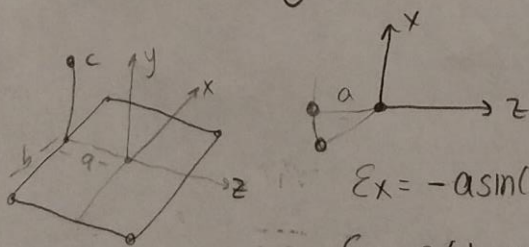
1. Roll errors, ϵ_x



$$G_z = Q \sin(\epsilon_x) = Q \epsilon_x$$

$$G_y = P \sin(\epsilon_x) = P \epsilon_x$$

2. Pitch errors, ϵ_y



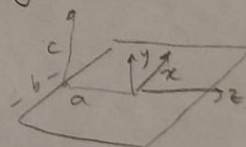
$$\epsilon_x = -a \sin(\epsilon_y) = -a$$

$$\epsilon_z = a(1 - \cos(\epsilon_y)) = \frac{a \epsilon_y^2}{2} = \frac{a \epsilon_y}{2}$$

ERROR GAINS

	G_x	G_y	G_z
ϵ_x	0	P	Q
ϵ_y	-P	0	$\frac{P \epsilon_y}{2}$
ϵ_z	-Q	$-\frac{Q \epsilon_z}{2}$	0

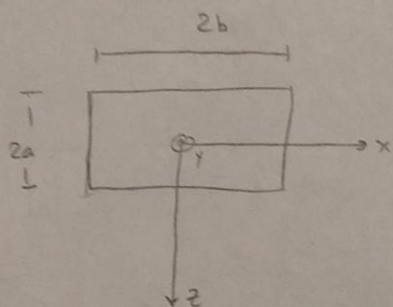
3. Yaw errors, ϵ_z



$$\epsilon_x = -c \sin(\epsilon_z)$$

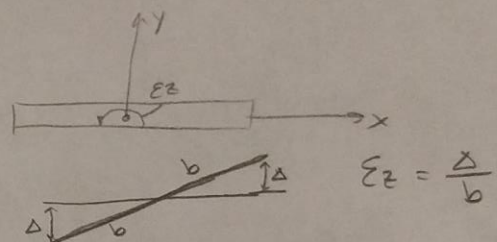
$$\epsilon_y = -c(1 - \cos(\epsilon_z))$$

Where do we get actual values for $\epsilon_x, \epsilon_y, \epsilon_z$?



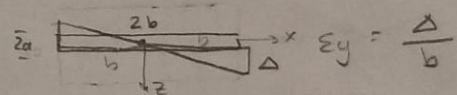
Δ is either running parallelism
or stop in rails from clearance gap

1. ϵ_z



if you are using preload,
these values ($\epsilon_z, \epsilon_y, \epsilon_x$)
can be substituted by the running
parallelism.

2. ϵ_y



3. ϵ_x

